

# ANUPREET SINGH

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## EDUCATION

### University of Maryland, Baltimore County

(Master of Science in Computer Science, GPA: 3.8/4.0)

Baltimore, MD

Aug 2024 – May 2026

### Indian Institute of Information Technology, Sonapat

(Bachelor of Technology in Computer Science and Engineering, GPA: 8.45/10)

Haryana, India

Aug 2019 – May 2023

## SKILLS / RELEVANT COURSEWORK

**Programming Languages:** C++, Python, JavaScript, SQL

**Tools and Frameworks:** CMake, Clang, LLVM, OpenGL, PyTorch, JAX, LangChain, Pinecone, FastAPI, PostgreSQL, GoogleTest, AWS, Redis, Kafka, Docker, Kubernetes, GitHub Actions(CI/CD)

**Areas of Interest and Coursework:** Systems Programming, AI & LLMs, Data Structures & Algorithms

## ACADEMIC EXPERIENCE / PROJECTS

### Graduate Assistant | UMBC - Operating Systems & Data Structures

Jan 2025 – June 2026

- Designed over 10 C++ assignments on compiler optimizations (like constant folding, DCE, loop unrolling) and OS fundamentals like concurrency, multithreading, and virtual memory (MMU address translation, multi-level page tables, and kernel page-fault handling) and graded coursework in computer architecture for 40+ students.
- Developed 200+ unit tests for the DSA projects on Linked Lists, Self Balancing Trees, Heaps, and Hash Tables automating logic checking and reducing grading time by over 80%.
- Helped 50+ students design and debug C++ projects implementing core data structures, and graded their homework.
- Held office hours to clarify technical concepts and translate requirements into actionable implementation steps.
- Mentored 10 student Scrum teams through sprint planning, reviews, and backlog refinement for their projects.
- Coordinated between 40 students and stakeholders, keeping all 10 projects on schedule amid evolving requirements.

### Rendering Pipelines from Scratch | Ray tracer, rasterizer, and real-time OpenGL renderer in C++

- Built a C++ ray tracer that parses Polyray scenes and shades them with Blinn-Phong direct lighting, recursive reflection, and Schlick-Fresnel refraction, then accelerated it with a median-split BVH over primitive centroids and an atomic row-chunk thread pool, cutting a 512x512 render from 106 s to 0.14 s (~759x).
- Implemented DXT1 texture compression end to end in an OpenGL scene renderer, fitting each 4x4 texel block's palette by principal-axis covariance, packing 2-bit selectors into the 64-bit block layout, and compressing every CPU-generated mip level independently, shrinking color textures 6x in GPU memory behind a runtime A/B toggle.

## INDUSTRY EXPERIENCE

### Data AI Engineer | Platinum Business Services (R&D Team) | Catonsville, MD

Dec 2025 – Present

- Built a priority-ordered fallback in the RAG chat agent across 5 embedding models (OpenAI, Anthropic, Ollama), re-embedding and re-running low-citation queries against each successive model's vector index to exploit differences in embedding representation and surface SSP evidence missed by higher-priority indexes, raising evidence-backed control assessment accuracy from 64% to 92%.
- Diagnosed an out-of-memory failure in that fallback path, tracing it to freshly pulled 4-5 GB model weights lingering in the OS page cache, and fixed it with targeted posix\_fadvise eviction, eliminating model-load failures.
- Developed backend services for an AI-native GRC platform that audits an SSP draft for compliance with NIST 800-53, chunking ~1,200 controls into a per-model vector index with linkage metadata tying each chunk to its parent control and sibling clauses, cutting a control-by-control review from days of manual work to <10 minutes.

### Software Development Engineer | Turfco Landscape Co. | Surrey, BC

Jan 2023 – June 2024

- Architected a self-serve quoting platform that priced jobs with deterministic multi-stage evaluation, JSONB encoded conditions, per-quote audit trails of every rule applied, and a write-invalidated Redis cache, all driven by rules editable from an admin console instead of a developer deploy, cutting quote turnaround from ~2 days to <1 min.
- Built the online ordering and booking platform by taking self-serve orders through Stripe Checkout deposits, confirming them with signature-verified idempotent webhooks, and relaying a transactional outbox to Kafka workers for order confirmation, customer notification, and crew allocation, taking \$600K+ in online bookings across ~1,100 orders.
- Designed and deployed the AWS architecture that took the business off a WordPress brochure site, with SPA on CloudFront, autoscaling Express tier behind an ALB across two AZs, Multi-AZ RDS, sized for ~10x seasonal peaks and load-tested to ~50 quote requests/sec at <120 ms p99 with zero-downtime rolling deploys.